

OL P1 -MOCK 2025

Computer Science 2210/0478



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Paper 1

1 A student is using a smartphone.

(a) Identify two input devices built into the smartphone.

1

2

[2]

(b) Identify one output device built into the smartphone.

.....

[1]

(c) Identify one type of storage device built into the smartphone.

.....

[1]

2 Hypertext markup language (HTML) colour codes can be represented as hexadecimal.

(a) Tick (✓) **three** boxes to show which statements about the hexadecimal number system are correct.

A It uses the values 0 to 9 and A to F.

☐

B It can be used as a shorter representation of binary.

☐

C It is a base 10 system.

☐

D It can be used to represent error codes.

☐

(b) Denary numbers can be converted to hexadecimal.

Convert the **three** denary numbers to hexadecimal.

20

32

165

[3]





Paper 1

3 The binary number 10100011 is stored in random access memory (RAM).

A logical left shift of **three** places is performed on the binary number.

(a) Give the 8-bit binary number that will be stored after the shift has taken place.

..... [1]

(b) Tick (✓) **one** box to show which statement about a logical left shift of **two** places is correct.

A It would divide the binary number by 2.

☐

B It would multiply the binary number by 2.

☐

C It would divide the binary number by 4.

☐

D It would multiply the binary number by 4.

☐

(c) 10100011 can be stored as a two's complement integer.

Convert the two's complement integer 10100011 to denary. Show all your working.

.....
.....
.....
..... [2]

(d) The binary number is measured as a byte because it has 8 bits.

State how many bytes there are in a kibibyte (KiB).

..... [1]



Paper 1

4 A school sends student data from one computer to another using a network.

(a) What is meant by the term data packet?

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.....

[2]

(b) The data is sent using packet switching.

Explain how packet switching works.

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.....
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.....
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.....

[3]

(c) The data is sent using serial transmission.

Give two reasons why serial transmission is used instead of parallel transmission.

- 1.
- 2.

[2]



Paper 1

5 A tablet computer uses memory and storage to work with programs and data.

(a) What is the purpose of **RAM** in a computer?

.....
.....

[2]

(b) The tablet also uses **secondary storage**.

Explain why secondary storage is needed in a computer.

.....
.....

[2]

(c) Tick (✓) **three** devices that are examples of **secondary storage**.

- ☐ Random Access Memory (RAM)
- ☐ Compact Disk (CD)
- ☐ Blu-ray Disk
- ☐ Read Only Memory (ROM)
- ☐ Hard Disk Drive (HDD)
- ☐ Graphics Card

[3]

(d) The processor uses the **fetch–decode–execute (FDE) cycle** to run programs.

Describe what happens in the **cycle**.

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.....
.....

[4]

6 A computer needs firmware and system software to operate.

(a) State the purpose of firmware.

.....
.....

[1]

(b) Give **one** example of firmware.

.....

[1]



Paper 1

(c) Give **two** examples of system software.

- 1
- 2 [2]

7 A company sends secure messages using encryption.
Complete the paragraph about **asymmetric encryption**.
Choose words from the list. Some words will not be used. Use each correct word only once.

asymmetric cipher text encrypted key plain text

private key public key decrypted certificate symmetric

The sender starts with some
This data is using the receiver's
.....

Only the receiver can turn it back into plain text by using their [4]

8 A factory uses an automated robot to move boxes from one conveyer belt to another.

(a) What does it mean when a robot is described as **automated**?

-
- [1]

(b) List **three characteristics** of an automated robot.

1.
2.
3. [3]

(c) The robot moves boxes and must stop when it reaches the end of the conveyer belt.
It uses **sensors** and a **microprocessor** to detect the end.

Paper 1

Explain how the robot uses the sensors and microprocessor to perform this task.

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[6]

(d) Give **two** advantages of using an automated robot to move boxes in a factory.

1.

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2.

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.....

[2]

Paper 1

(e) Give **two** disadvantages of using an automated robot in a factory.

1.
.....
.....
2.
.....
.....

[2]

(f) The robot is upgraded to use machine learning.
Explain how this could improve the robot’s performance.

.....
.....
.....

[2]

9 A company owner has installed a new network. Data is correct before it is transmitted across the network.

The company owner is concerned that data might have errors after transmission.

(a) Explain how the data might have errors after transmission.

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.....
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.....

[3]

Paper 1

(c) The company owner also installs a firewall to help protect the network from hackers and malware.

(i) Explain how the firewall operates to help protect the network.

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[5]

(ii) Give **two** examples of malware that the firewall can help protect the network from and how the malware work.

1

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2

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[4]